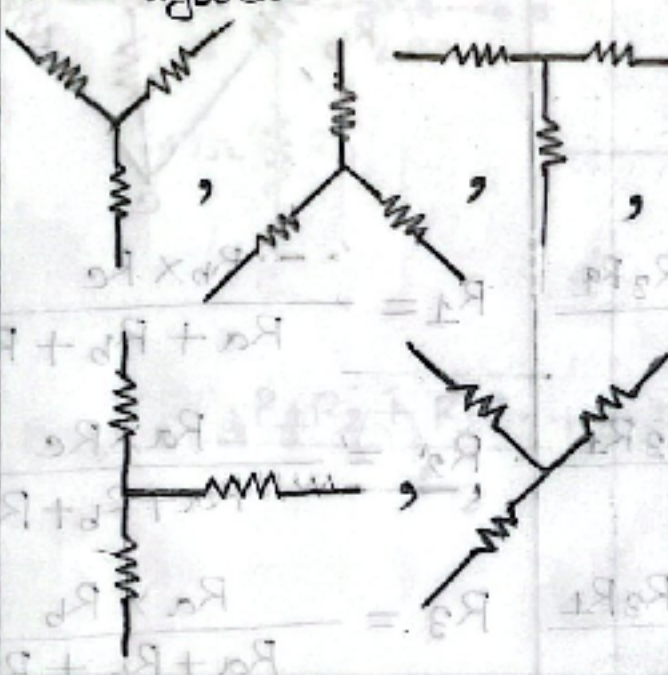
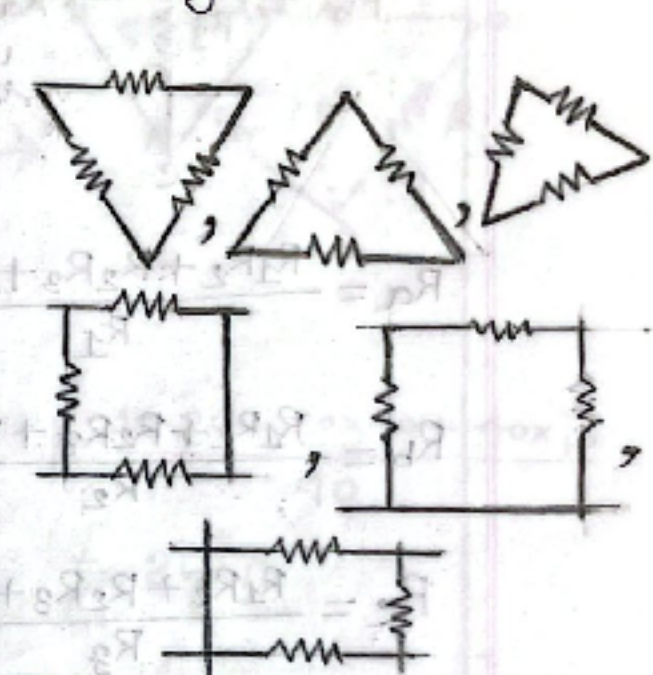
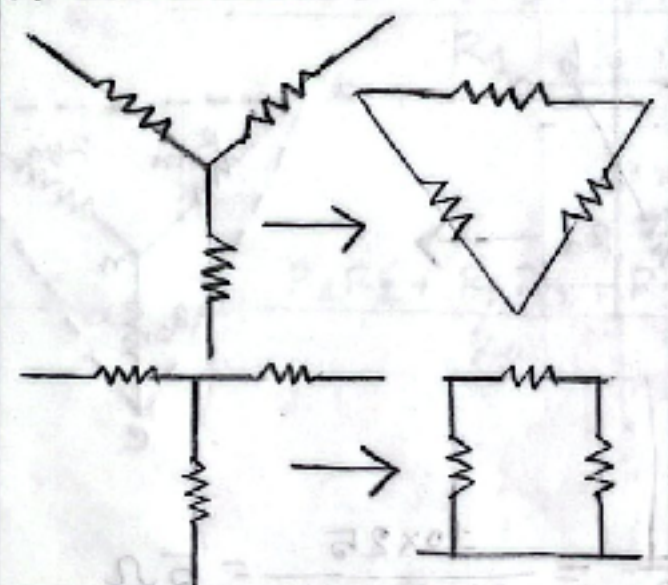
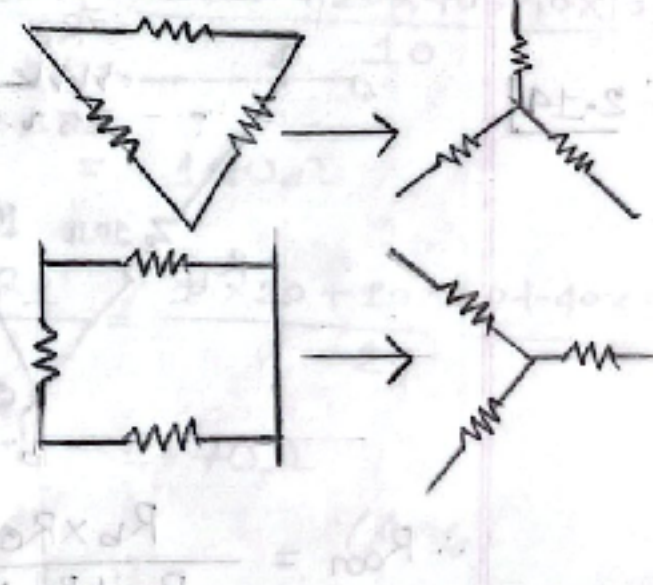
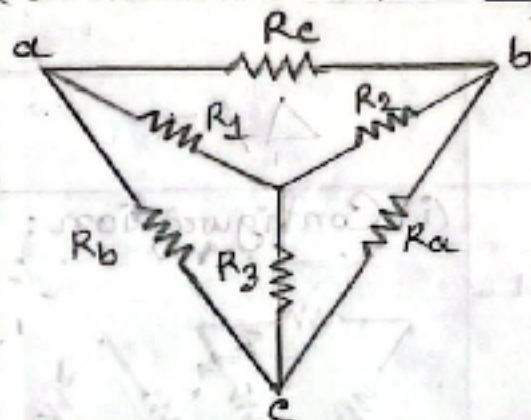


WYE-Delta Transformation -

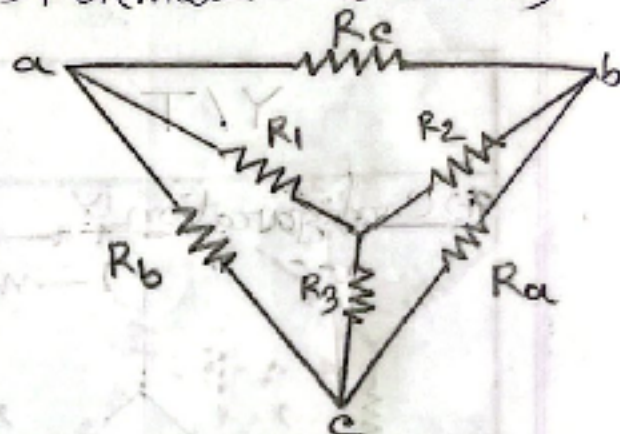
Y/T	Δ/π
(i) Configuration : 	(i) Configuration : 
(ii) Conversion : 	(ii) Conversion : 
(iii) Looks like the letter Y/T.	(iii) Looks like a triangle (Δ)/π.
(iv) Three resistors connected to a common point.	(iv) Three resistors connected in a closed loop.

(v) Formula: ($Y \rightarrow \Delta$)

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

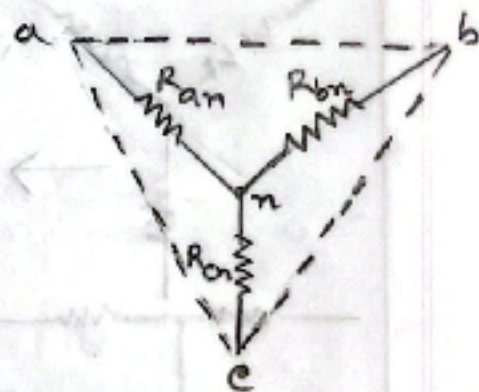
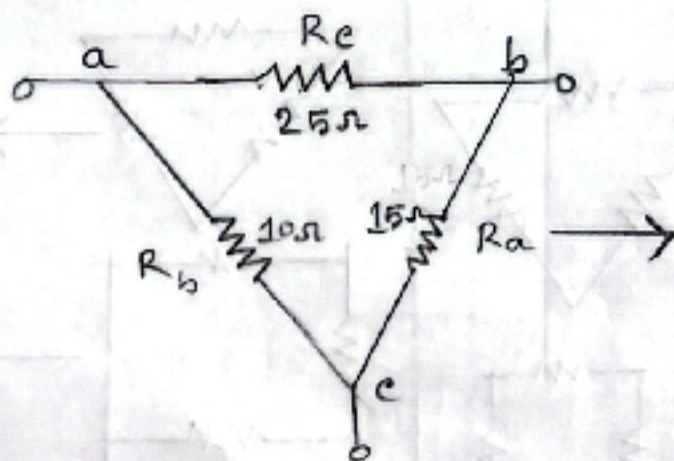
(v) Formula: ($\Delta \rightarrow Y$)

$$R_1 = \frac{R_b \times R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a \times R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a \times R_b}{R_a + R_b + R_c}$$

2.14

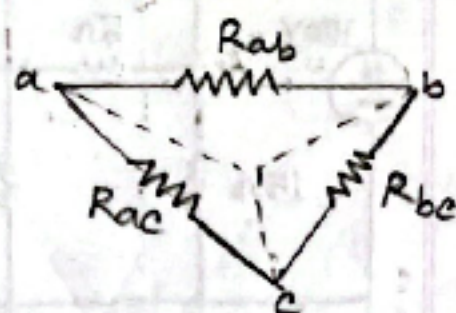
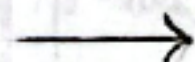
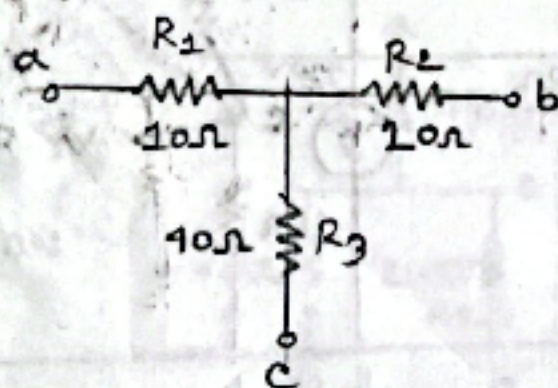


$$\therefore R_{an} = \frac{R_b \times R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{10 + 15 + 25} = 5 \Omega$$

$$\therefore R_{bn} = \frac{R_a \times R_c}{R_a + R_b + R_c} = \frac{15 \times 25}{10 + 15 + 25} = 7.5 \Omega$$

$$\therefore R_{cn} = \frac{R_b \times R_a}{R_a + R_b + R_c} = \frac{10 \times 15}{10 + 15 + 25} = 3 \Omega \quad (\text{Ans.})$$

P-2.14

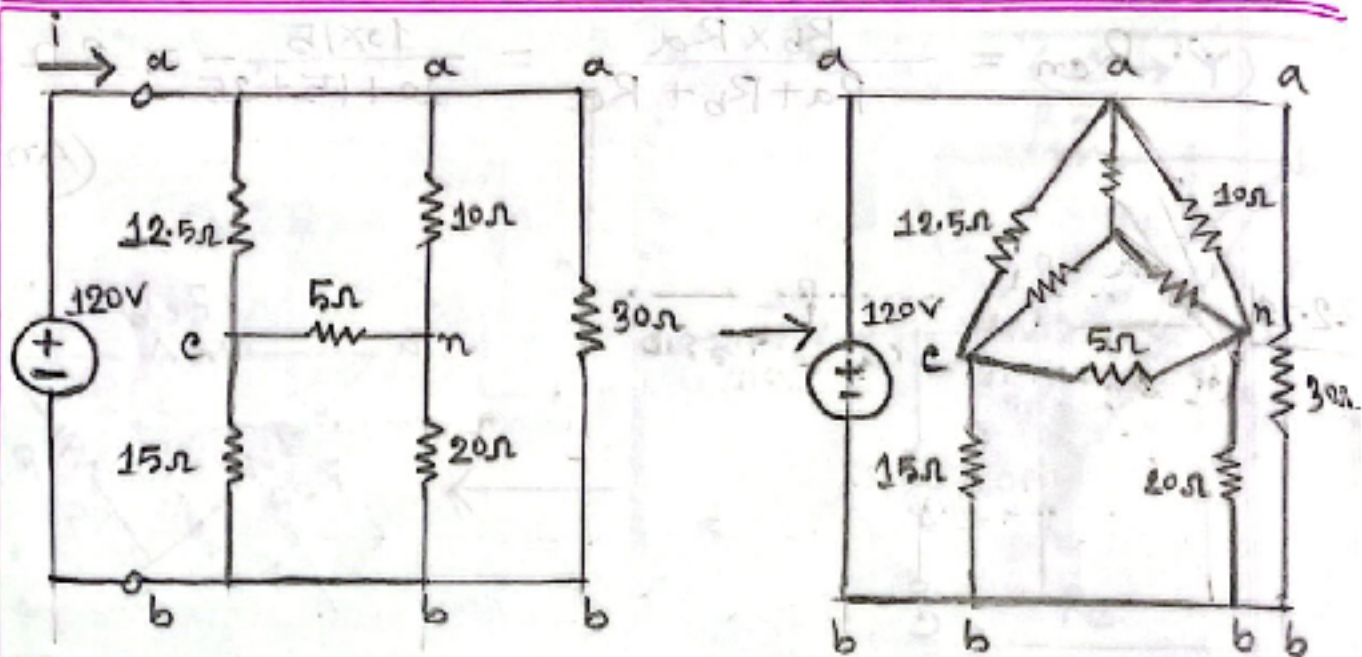


$$\therefore R_{ab} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = \frac{10 \times 20 + 20 \times 40 + 40 \times 10}{40} = 35 \Omega$$

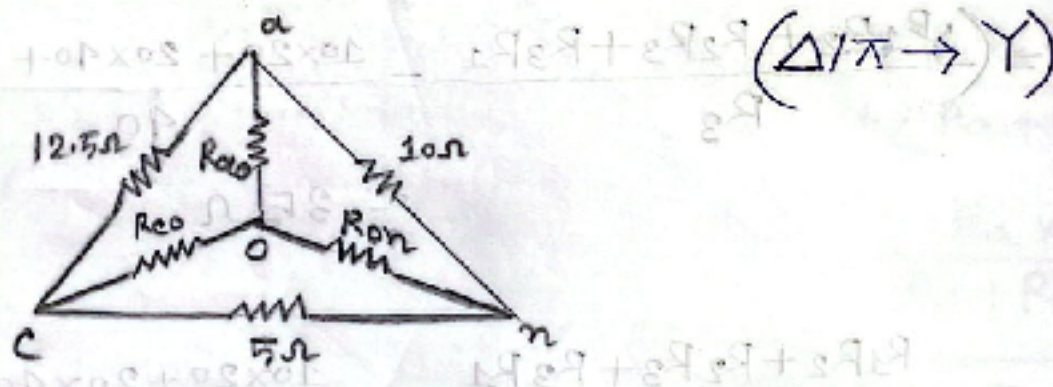
$$\therefore R_{bc} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} = \frac{10 \times 20 + 20 \times 40 + 40 \times 10}{10} = 140 \Omega$$

$$\therefore R_{ac} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} = \frac{10 \times 20 + 20 \times 40 + 40 \times 10}{20} = 70 \Omega \quad (\text{Ans.})$$

2.15



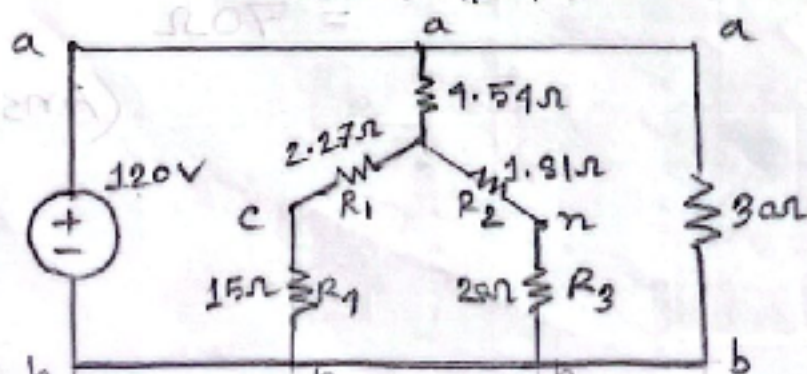
Find R_{ab} and i .



$$\therefore R_{ao} = \frac{12.5 \times 10}{12.5 + 10 + 5} = 4.54 \Omega$$

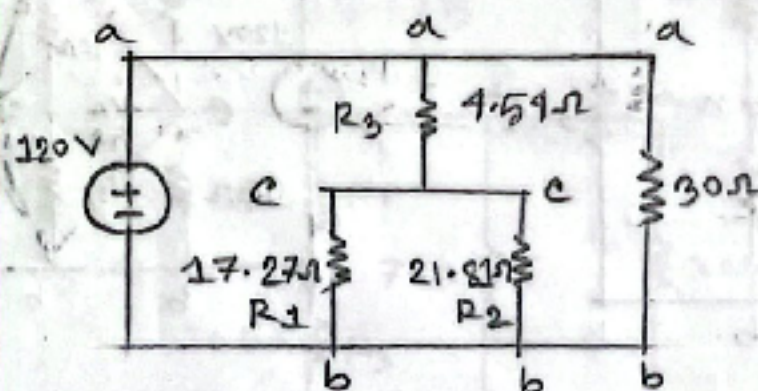
$$\therefore R_{co} = \frac{12.5 \times 5}{12.5 + 10 + 5} = 2.27 \Omega$$

$$\therefore R_{on} = \frac{10 \times 5}{12.5 + 10 + 5} = 1.81 \Omega$$



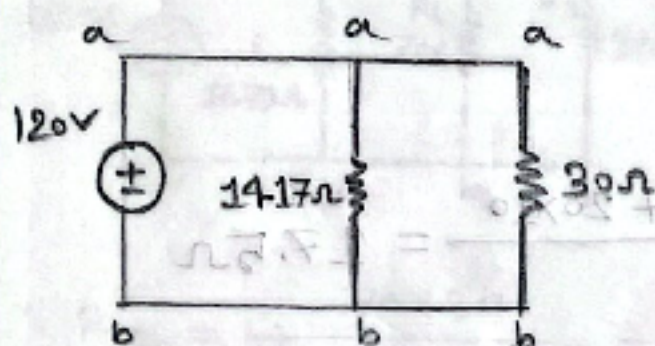
$$\therefore R_1 = 2.27 + 15 = 17.27 \Omega \quad (\Delta \rightarrow \Delta \text{ HY})$$

$$\therefore R_2 = 1.81 + 20 = 21.81 \Omega$$



$$\therefore R_1 = \frac{17.27 \times 21.81}{17.27 + 21.81} = 9.63 \Omega$$

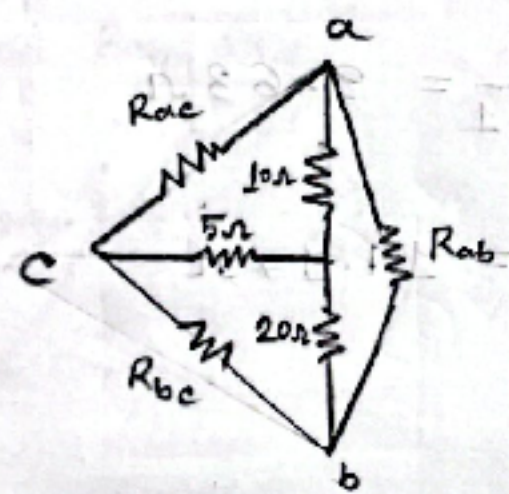
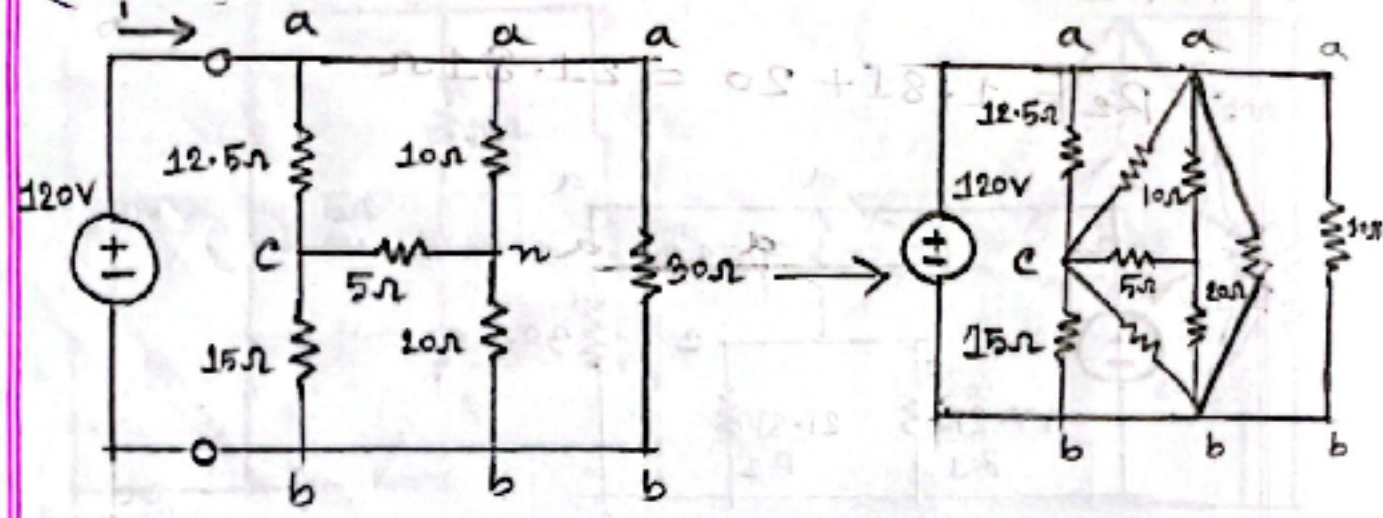
$$\therefore R_3 = 4.54 + 9.63 = 14.17 \Omega$$



$$\therefore R_{ab} = \frac{30 \times 14.17}{30 + 14.17} = 9.62 \Omega$$

$$\therefore i = \frac{V}{R_{ab}} = \frac{120}{9.62} = 12.47 \text{ A} \quad (\text{Ans.})$$

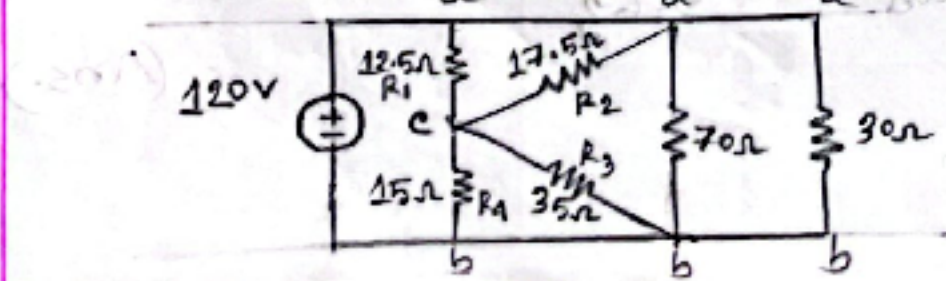
(Y/T → Δ)



$$\therefore R_{ac} = \frac{10 \times 5 + 5 \times 20 + 20 \times 10}{20} = 17.5 \Omega$$

$$\therefore R_{bc} = \frac{10 \times 5 + 5 \times 20 + 20 \times 10}{10} = 35 \Omega$$

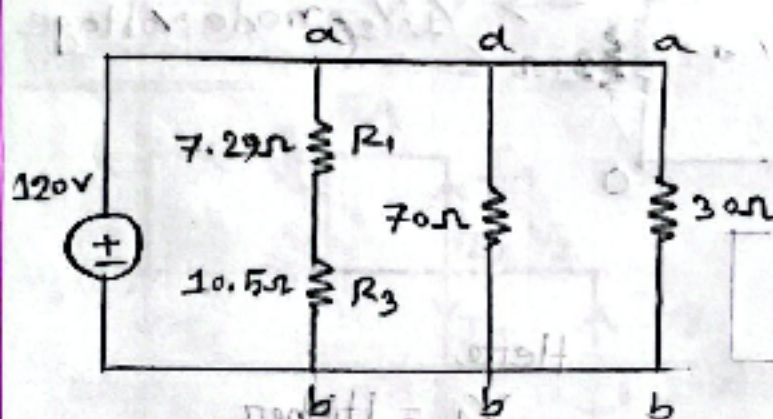
$$\therefore R_{ab} = \frac{10 \times 5 + 20 \times 5 + 20 \times 10}{5} = 70 \Omega$$



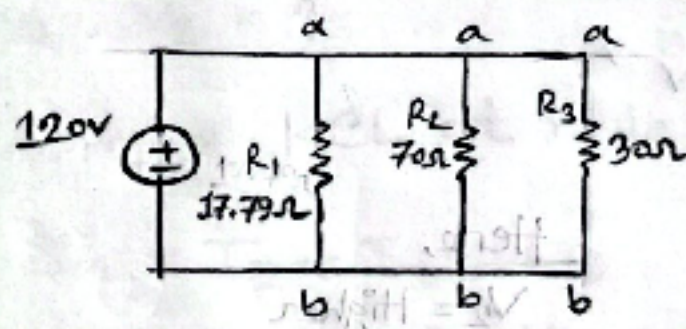
Example 10

$$R_1 = \frac{12.5 \times 17.5}{12.5 + 17.5} = 7.29 \Omega$$

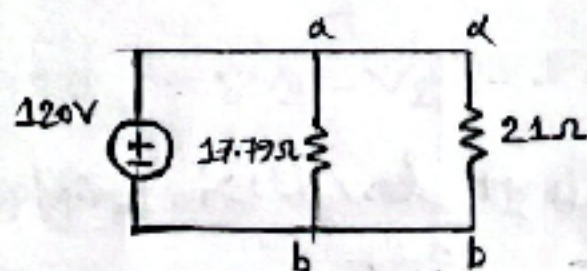
$$R_3 = \frac{35 \times 15}{35 + 15} = 10.5 \Omega$$



$$\therefore R_1 = 7.29 + 10.5 = 17.79 \Omega$$



$$\therefore R_2 = \frac{70 \times 30}{70 + 30} = 21 \Omega$$

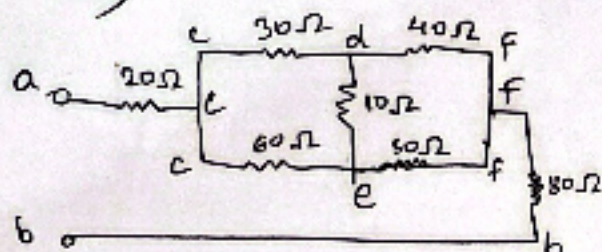


$$\therefore R_{ab} = \frac{21 \times 17.79}{17.79 + 21} = 9.62 \Omega$$

$$\therefore i = \frac{V}{R_{ab}} = \frac{120}{9.62} = 12.47 \text{ A (Ans.)}$$

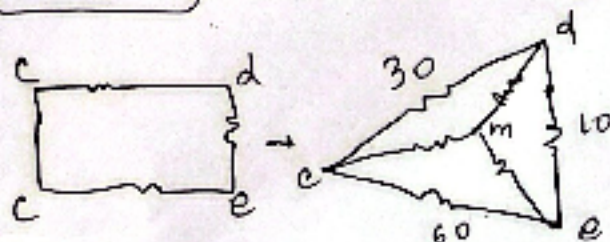
2.53

a)



Q: Obtain equivalent resistance R_{ab}

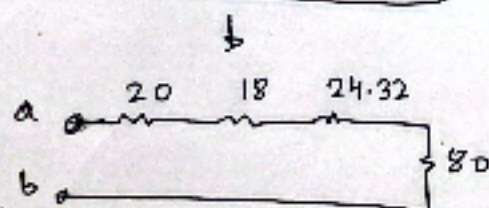
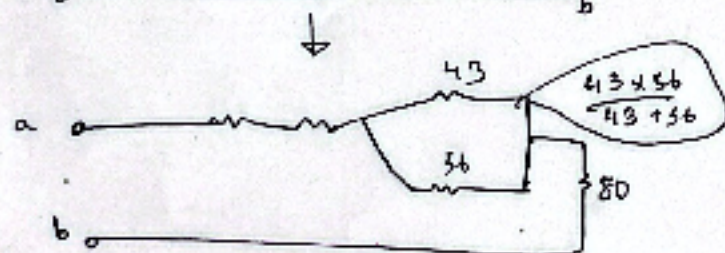
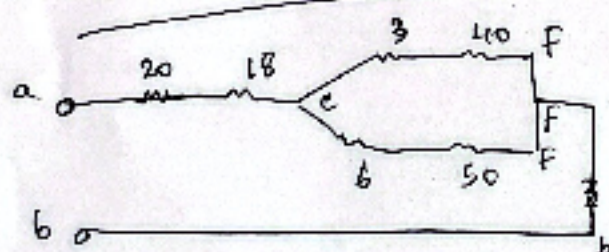
$\Delta \rightarrow Y$



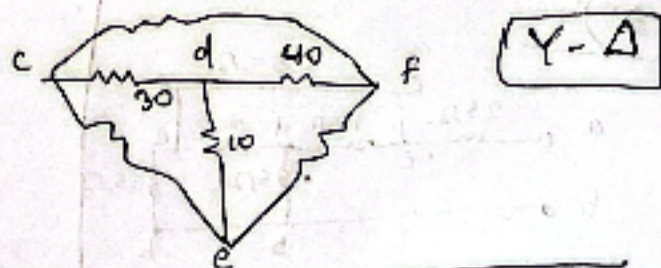
$$R_{cm} = \frac{30 \times 60}{30 + 60 + 10} = 18 \Omega$$

$$R_{dm} = \frac{30 \times 10}{100} = 3 \Omega$$

$$R_{em} = \frac{10 \times 60}{100} = 6 \Omega$$



$$E R_{ab} = 20 + 18 + 2.43 + 80 = 142.32 \Omega$$



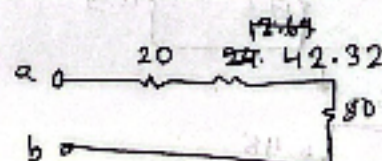
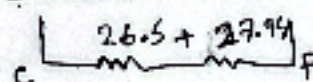
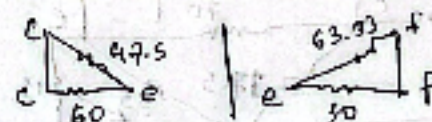
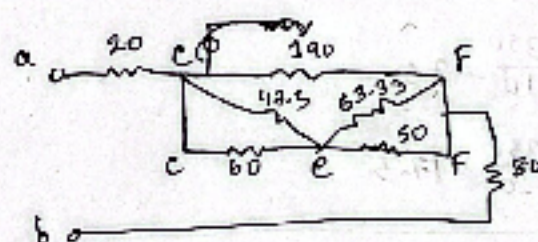
$$R_{cf} = \frac{30 \times 40 + 40 \times 10 + 10 \times 30}{10}$$

$$= 190$$

$$R_{af} = \frac{1900}{30} = 63.33$$

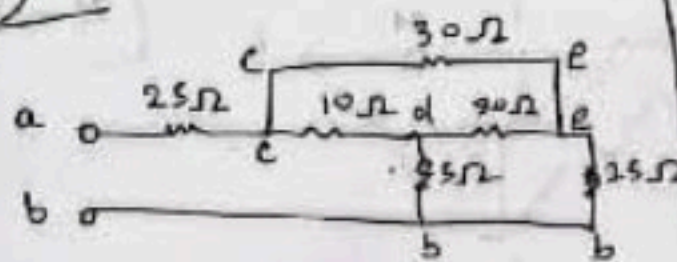
$$R_{ee} = \frac{1900}{40}$$

$$= 47.5$$

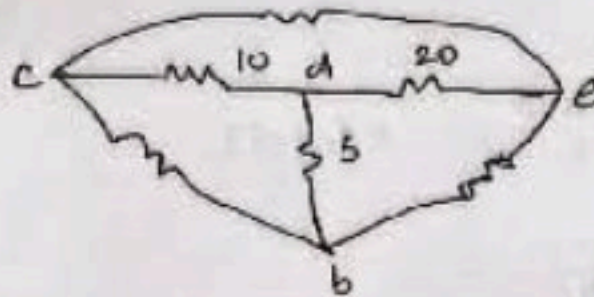


$$E R_{ab} = 80 + 42.32 + 20 = 142.32 \Omega$$

2.51)b.



$$ER_{ab} = ?$$

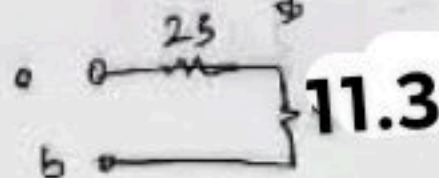
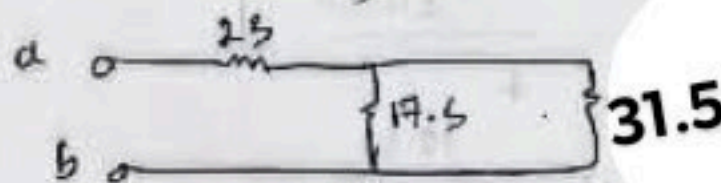
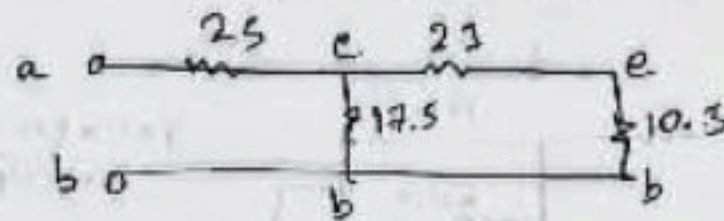
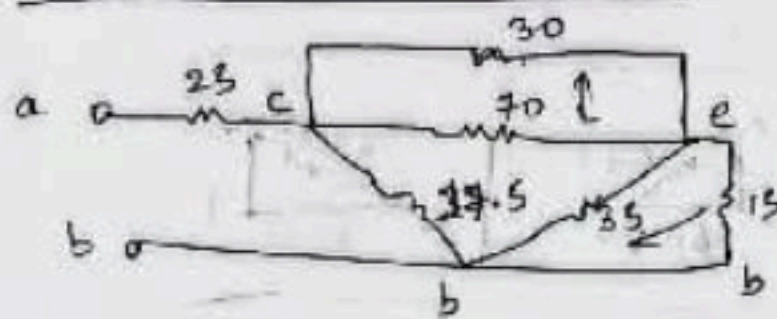


$$R_{ce} = \frac{10 \times 20 + 20 \times 5 + 5 \times 10}{5} = \frac{350}{5}$$

$$= 70$$

$$R_{eb} = \frac{350}{10} = 35$$

$$R_{cb} = \frac{350}{20} = 17.5$$



$$ER_{ab} = 36.3$$